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Objective

To pursue an MSc in Computer Science with a focus on Artificial Intelligence and applied machine learning. Aims to strengthen expertise in deep learning, computer vision, and intelligent systems while exploring interdisciplinary applications of AI. Intends to build a career in research and development within AI-driven solutions, working in roles such as Machine Learning Engineer or Applied AI Researcher after graduation.

Education

SIES Graduate School of Technology, BE in Artificial Intelligence and Machine Learning 2022 – 2026

- GPA: 8.76/10
- **Coursework:** Computer Vision, NLP, Data Visualization, Web Technologies, AI/ML.

Projects

SETU – Dual Coding Interoperability API Sep 2025
Team Size: 6

- Built SETU, a FHIR-compliant API using Bi-Encoder + Cross-Encoder pipeline for integrating NAMASTE (Ayush) and ICD-11 codes, improving interoperability in EMRs, insurance claims, and clinical decision support.
- **Tools/Technologies:** Python, FastAPI, Sentence-Transformers, FAISS, Bi-Encoder: SapBERT, Cross Encoder, FHIR R4, ABHA compliance, API microservice using AWS.

SeaGuard – Fishermen Safety & Navigation App Sep 2025
Team Size: 5

- Designed and developed a mobile-first app to improve maritime safety by helping fishermen track weather conditions, navigation, and SOS alerts; won HackCelestial 2.0 and was selected for a science exhibition under the Chairman of ISRO.
- **GitHub Repository:** github.com/Shucrates/SeaGuard
- **Tools/Technologies:** React Native with Expo, Twilio Integration, FastAPI, Mistral Saba and Sarvam AI for multilingual support, Custom Components & Stylesheets, Figma.

AutoPharmaX – Cancer Drug Response Prediction July 2025
Team Size: 4

- Built an ML pipeline to predict cancer drug response (LN(IC50)) using GDSC cell-line and drug datasets, applying preprocessing, feature engineering, and model tuning.
- Developed a Streamlit web app for interactive predictions, integrating the best XGBoost model and enabling visualization of results for different drug–cell line pairs.
- **Tools/Technologies:** Python (pandas, NumPy, scikit-learn, XGBoost, LightGBM), Streamlit, FAISS, Matplotlib, Seaborn.
- **GitHub Repository:** github.com/Shucrates/autopharmax

Experience

Graphic & Web Designer Intern, Prepbook (Edtech)–Mumbai June 2025 – Aug 2025

Project/Role: Redesigned and improved the digital platforms of Prepbook, an e-learning company focused on UPSC and government exam preparation.

- Redesigned the company website, modernizing the interface and improving usability, which led to enhanced student engagement.

- Assisted in developing the core web application for learners, focusing on navigation, accessibility, and smooth functionality.
- Collaborated with the core team to align UI/UX design with product goals.
- Gained hands-on experience with Figma, frontend design, and user-focused product development.

Python Developer Intern, Ion Chemicals – Navi Mumbai

Jan 2025 – Apr 2025

Project/Role: Developed a computer vision model to improve anomaly detection in industrial pipelines.

- Built and optimized a model using Python and OpenCV to detect fluid and gas flow anomalies.
- Conducted real-time data analysis, contributing to more accurate monitoring of industrial systems.
- Collaborated with the engineering team to explore integration of the solution into existing workflows.
- Presented outcomes for internal review, supporting potential adoption of the project for future operations.

Design Intern, Irsya (Clothing Startup) – Mumbai

Sep 2023 – Jan 2024

Project/Role: Supported the initial launch phase of Irsya by working on branding, product design, and market positioning.

- Designed and contributed to the early designs of the apparel collections, aligning with the brand vision.
- Assisted in shaping the brand identity by working on marketing collateral, product aesthetics, and visual communication.
- Collaborated with the founding team to ensure consistency in brand image at all digital and offline touch points.
- Contributed to the successful launch of the startup's product, strengthening its initial presence in the market.

Publications/Research/White Papers

Attention Enhanced Speaker representation with Contrastive Learning

Developed a Transformer-based speaker encoder using contrastive (GE2E) learning to improve speaker verification and zero-shot TTS, achieving lower error rates and faster inference compared to LSTM baselines.

Positions of Responsibility

Joint Technical Secretary, Technical Team

SIES GST | 2023–2025

- Led the coordination of technical events and workshops, including hackathons, tech fests, and innovation challenges.
- Managed communications between the technical team and other departments, ensuring smooth project execution.
- Provided technical guidance to students, fostering skill development and active participation within the organization.

Junior Head, Aaroha Music Club

SIES GST | 2023–2025

- Directed musical events, performances, and practice sessions while collaborating with club members to organize cultural activities and competitions.
- Promoted music culture on campus by encouraging student creativity and participation.
- Assisted in resource management and coordinated event planning with senior members to ensure successful execution.

Technical Skills

Languages: Python, JavaScript, HTML, CSS, React Native

Technologies: FastAPI, React Native with Expo, PyTorch, YOLOv8, Scikit-learn, XGBoost, OpenCV, NumPy, Pandas, Streamlit, Sentence-Transformers, Figma, Adobe Suite

Languages

Languages: English (proficient), Hindi (fluent), Marathi (native)